

PRACTICAL LABORATORY ASSIGNMENT

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Course	GRC 102 Week 4
Linux Distribution Used	Kali Linux

Module 1 System Auditing with auditd

Activity 1.1 Verify or Install auditd

```
(icdfa@icdfa)-[~]
$ cd /etc
$ sudo systemctl status auditd
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset:
   Active: active (running) since Fri 2026-10-02 07:23:49 WAT; 9min ago
     Invocation: 226cf0a42d6345e38fc7261eaaa7f249
       Docs: man:auditd(8)
             https://github.com/linux-audit/audit-documentation|
    Main PID: 19210 (auditd)
      Tasks: 2 (limit: 2207)
     Memory: 680K (peak: 1.7M)
        CPU: 45ms
      CGroup: /system.slice/auditd.service
              └─19210 /usr/sbin/auditd

Oct 02 07:23:49 icdfa systemd[1]: Starting auditd.service - Security Audit L
Oct 02 07:23:49 icdfa auditd[19210]: No plugins found, not dispatching events
Oct 02 07:23:49 icdfa auditd[19210]: Init complete, auditd 4.1.2 listening f
Oct 02 07:23:49 icdfa systemd[1]: Started auditd.service - Security Audit Lo
lines 1-17/17 (END)... skipping ...
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-10-02 07:23:49 WAT; 9min ago
     Invocation: 226cf0a42d6345e38fc7261eaaa7f249
       Docs: man:auditd(8)
             https://github.com/linux-audit/audit-documentation
    Main PID: 19210 (auditd)
      Tasks: 2 (limit: 2207)
     Memory: 680K (peak: 1.7M)
        CPU: 45ms
      CGroup: /system.slice/auditd.service
              └─19210 /usr/sbin/auditd

Oct 02 07:23:49 icdfa systemd[1]: Starting auditd.service - Security Audit Logging Service ...
Oct 02 07:23:49 icdfa auditd[19210]: No plugins found, not dispatching events
Oct 02 07:23:49 icdfa auditd[19210]: Init complete, auditd 4.1.2 listening for events (startup state enable)
Oct 02 07:23:49 icdfa systemd[1]: Started auditd.service - Security Audit Logging Service.
~
```

Figure 1 Terminal showing auditd as active/running.

Activity 1.2 Configure Audit Rules

```
(icdfa@icdfa)-[~]
$ _w /etc/passwd -p rwx -k passwd_changes
-w: command not found

(icdfa@icdfa)-[~]
$ _w /etc/passwd -p rwx -k shadow_changes
-w: command not found

(icdfa@icdfa)-[~]
$
$ sudo nano /etc/audit/rules.d/custom.rules
[sudo] password for icdfa:

(icdfa@icdfa)-[~]
$ sudo systemctl restart auditd

(icdfa@icdfa)-[~]
$ sudo auditctl -l
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -F key=program_execution

(icdfa@icdfa)-[~]
$
```

Figure 2 Terminal output showing loaded custom rules.

Activity 1.3 Generate and Query Audit Events

1.3.1

```
(icdfa@icdfa)-[~]
$ sudo ausearch -k passwd_changes

time→Fri Oct 2 08:05:34 2026
type=PROCTITLE msg=audit(1790924734.146:157): proctitle=2F7362696E2F61756469746
3746C002D52002F6574632F61756469742F61756469742E72756C6573
type=SYSCALL msg=audit(1790924734.146:157): arch=c000003e syscall=44 success=ye
s exit=1084 a0=3 a1=7ffff0c891f0 a2=43c a3=0 items=0 ppid=42481 pid=42500 auid=
4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=(none) s
es=4294967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(nul
l)
type=CONFIG_CHANGE msg=audit(1790924734.146:157): auid=4294967295 ses=429496729
5 subj=unconfined op=add_rule key="passwd_changes" list=4 res=1

time→Fri Oct 2 08:05:53 2026
type=PROCTITLE msg=audit(1790924753.886:282): proctitle=7375646F006175646974637
46C002D6C
type=PATH msg=audit(1790924753.886:282): item=0 name="/etc/passwd" inode=110558
7 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790924753.886:282): cwd="/home/icdfa"
type=SYSCALL msg=audit(1790924753.886:282): arch=c000003e syscall=257 success=y
es exit=3 a0=fffff9c a1=7f6b547e0d7f a2=80000 a3=0 items=1 ppid=15017 pid=4269
2 auid=1000 uid=1000 gid=1000 euid=0 suid=0 fsuid=0 egid=1000 sgid=1000 fsgid=1
000 tty=pts0 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_
changes"

time→Fri Oct 2 08:05:53 2026
type=PROCTITLE msg=audit(1790924753.890:283): proctitle=7375646F006175646974637
46C002D6C
type=PATH msg=audit(1790924753.890:283): item=0 name="/etc/passwd" inode=110558
7 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790924753.890:283): cwd="/home/icdfa"
type=SYSCALL msg=audit(1790924753.890:283): arch=c000003e syscall=257 success=y
es exit=5 a0=fffff9c a1=7f6b547e0d7f a2=80000 a3=0 items=1 ppid=15017 pid=4269
2 auid=1000 uid=1000 gid=1000 euid=0 suid=0 fsuid=0 egid=1000 sgid=1000 fsgid=1
000 tty=pts0 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_
changes"

time→Fri Oct 2 08:05:53 2026
type=PROCTITLE msg=audit(1790924753.898:284): proctitle=7375646F006175646974637
46C002D6C
type=PATH msg=audit(1790924753.898:284): item=0 name="/etc/passwd" inode=110558
7 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
```

Figure 3 Results of ausearch for a configured key

1.3.2

```
(icdfa@icdfa)-[~]
$ sudo aureport
Summary Report
=====
Range of time in logs: 02/10/2026 07:23:49.615 - 02/10/2026 08:15:34.182
Selected time for report: 02/10/2026 07:23:49 - 02/10/2026 08:15:34.182
Number of changes in configuration: 9
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 1
Number of failed authentications: 3
Number of users: 3
Number of terminals: 7
Number of host names: 2
Number of executables: 22
Number of commands: 26
Number of files: 27
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 2
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 3
Number of process IDs: 3664
Number of events: 3940
```

Figure 4 Summary output from aureport

Why This Event Matters to Security Governance

1. **Traceability & Non-Repudiation:** Connects system-level changes directly to an individual user identity rather than a generic shared account, fulfilling compliance requirements for user accountability.
2. **Unauthorized Access Detection:** Any unexpected write activity on core account files (/etc/passwd, /etc/shadow) serves as an early indicator of unauthorized privilege escalation or malicious backdoor account creation.
3. **Continuous Control Monitoring:** Proves to internal auditors and oversight bodies that security controls monitoring privileged file integrity are active, functional, and capturing evidence in real time.

Module 2 Log Management and Analysis

Activity 2.1 - Explore Logs with journalctl

```
(icdfa@icdfa)-[~]
$ sudo journalctl
[sudo] password for icdfa:
May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (dev@kali.org) (x86_64)
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000000000]
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x0000000000000000]
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x0000000000000000]
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000001000000-0x0000000000007fff]
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000007fff000-0x0000000000007fff]
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00000]
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00000]
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fff00000-0x00000000fff00000]
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS V1.0
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
May 10 20:50:54 icdfa kernel: Hypervisor detected: KVM
May 10 20:50:54 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
May 10 20:50:54 icdfa kernel: kvm-clock: using sched offset of 155397490851 cycles
May 10 20:50:54 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff
May 10 20:50:54 icdfa kernel: tsc: Detected 2419.196 Mhz processor
May 10 20:50:54 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable
May 10 20:50:54 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
May 10 20:50:54 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
May 10 20:50:54 icdfa kernel: MTRR map: 4 entries (3 fixed + 1 variable; max 32 entries)
May 10 20:50:54 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB
May 10 20:50:54 icdfa kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
May 10 20:50:54 icdfa kernel: RAMDISK: [mem 0x282fb000-0x30174fff]
May 10 20:50:54 icdfa kernel: ACPI: Early table checksum verification disabled
May 10 20:50:54 icdfa kernel: ACPI: RSDP 0x0000000000000000 000024 (v02 VBOX )
May 10 20:50:54 icdfa kernel: ACPI: XSDT 0x000000007fff0030 00003c (v01 VBOX )
May 10 20:50:54 icdfa kernel: ACPI: FACP 0x000000007fff00f0 0000f4 (v04 VBOX )
May 10 20:50:54 icdfa kernel: ACPI: DSDT 0x000000007fff0610 002353 (v02 VBOX )
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007fff0200 000040
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007fff0200 000040
May 10 20:50:54 icdfa kernel: ACPI: APIC 0x000000007fff0240 000054 (v02 VBOX )
May 10 20:50:54 icdfa kernel: ACPI: SSDT 0x000000007fff02a0 00036c (v01 VBOX )
May 10 20:50:54 icdfa kernel: ACPI: Reserving FACP table memory at [mem 0x7ffff000-0x7ffff000]
May 10 20:50:54 icdfa kernel: ACPI: Reserving DSDT table memory at [mem 0x7ffff000-0x7ffff000]
May 10 20:50:54 icdfa kernel: ACPI: Reserving FACS table memory at [mem 0x7ffff000-0x7ffff000]
May 10 20:50:54 icdfa kernel: ACPI: Reserving FACS table memory at [mem 0x7ffff000-0x7ffff000]
May 10 20:50:54 icdfa kernel: ACPI: Reserving APIC table memory at [mem 0x7ffff000-0x7ffff000]
May 10 20:50:54 icdfa kernel: ACPI: Reserving SSDT table memory at [mem 0x7ffff000-0x7ffff000]
```

Figure 5 Terminal showing recent system log review using journalctl

Activity 2.2 - Authentication and Privilege-Use Analysis

```
(icdfa@icdfa)-[~]
$ sudo less /var/log/auth.log
/var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "failed password" /var/log/auth.log
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "sudo" /var/log/auth.log
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$
```

Figure 6 Authentication/privilege-use log search results

Activity 2.3 General System Log Review

```
(icdfa@icdfa)-[~]
$ sudo less /var/log/auth.log
/var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "failed password" /var/log/auth.log
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "sudo" /var/log/auth.log
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ sudo less /var/log/syslog
[sudo] password for icdfa:
/var/log/syslog: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "error" /var/log/syslog
grep: /var/log/syslog: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "warning" /var/log/syslog
grep: /var/log/syslog: No such file or directory
```

Figure 7 Errors or warnings found in syslog

Module 3 Linux Security Assessment with Lynis

Activity 3.1 Install or Access Lynis

3.1.1

```
Places
(icdfa@icdfa)-[~]
$ sudo .lynis audit system
[sudo] password for icdfa:
sudo: .lynis: command not found

(icdfa@icdfa)-[~]
$ sudo lynis audit system

[ Lynis 3.1.4 ]

#####
#
# Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
# welcome to redistribute it under the terms of the GNU General Public License.
# See the LICENSE file for details about using this software.
#
# 2007-2024, CISOfy - https://cisofy.com/lynis/
# Enterprise support available (compliance, plugins, interface and tools)
#####
# Devices

[+] Initializing program
-----
- Detecting OS ... [ DONE ]
- Checking profiles ... [ DONE ]

-----
Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.13
Hardware platform: x86_64
Hostname: icdfa

Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins
```

Figure 8 Lynis version and execution start screen

3.1.2

```
Lynis security scan details:

Hardening index : 65 [##### ]
Tests performed : 273
Plugins enabled : 1
```

Figure 9 Lynis completion screen displaying the Hardening Index

3.1.3

```
Warnings (1):
-----
! iptables module(s) loaded, but no rules active [FIRE-4512]
https://cisofy.com/lynis/controls/FIRE-4512/
```

Figure 10 Warning

3.1.4

```
Suggestions (40):
* This release is more than 4 months old. Check the website or GitHub to see
if there is an update available. [LYNIS]
- Related resources
  * Website: https://cisofy.com/lynis/controls/LYNIS/
* Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions [DEB-0280]
- Related resources
  * Website: https://cisofy.com/lynis/controls/DEB-0280/
* Install apt-listbugs to display a list of critical bugs prior to each APT i
nstallation. [DEB-0810]
- Related resources
  * Website: https://cisofy.com/lynis/controls/DEB-0810/
* Install apt-listchanges to display any significant changes prior to any upg
rade via APT. [DEB-0811]
- Related resources
  * Website: https://cisofy.com/lynis/controls/DEB-0811/
* Install needrestart, alternatively to debian-goodies, so that you can run n
eedrestart after upgrades to determine which daemons are using old versions of
libraries and need restarting. [DEB-0831]
- Related resources
  * Website: https://cisofy.com/lynis/controls/DEB-0831/
* Install fail2ban to automatically ban hosts that commit multiple authentica
tion errors. [DEB-0880]
- Related resources
  * Website: https://cisofy.com/lynis/controls/DEB-0880/
* Set a password on GRUB boot loader to prevent altering boot configuration (
e.g. boot in single user mode without password) [BOOT-5122]
- Related resources
  * Website: https://cisofy.com/lynis/controls/BOOT-5122/
* Determine runlevel and services at startup [BOOT-5180]
- Related resources
  * Website: https://cisofy.com/lynis/controls/BOOT-5180/
* Consider hardening system services [BOOT-5264]
```

Figure 11 Suggestion

Module 4 SIEM, Automation and Continuous Monitoring

1. Enterprise Integration (Forwarding & Representation)

To integrate your host-level evidence into an enterprise SIEM/SOAR environment:

Log Forwarding (auditd, journalctl, /var/log/auth.log):

- Use a log shipper (such as Filebeat, Fluentd, or rsyslog) installed on the endpoint to collect /var/log/audit/audit.log and /var/log/auth.log.
- Send the raw events over encrypted TLS (e.g., Syslog/TLS or Lumberjack protocol) to a central SIEM aggregator (like Splunk, Elastic SIEM, or Microsoft Sentinel).
- Systemd journal entries (journalctl) are captured directly via plugins (e.g., Filebeat's journald input module).

Log Parsing & Normalization:

- Raw auditd key-value pairs (like key="passwd_changes", type=SYSCALL, success=yes) are mapped to common taxonomy frameworks like Common Event Format (CEF) or Elastic Common Schema (ECS).
- This allows SOC analysts to query /etc/passwd modifications across thousands of Linux servers using standardized field names (e.g., file.path, event.action).

Audit Assessment Forwarding (Lynis):

- Lynis execution output (/var/log/lynis.log and /var/log/lynis-report.dat) is shipped after each automated scan.

- The Hardening Index score and specific Lynis Warning/Suggestion IDs are ingested as asset compliance metrics rather than high-frequency real-time alerts.

2. Technical Operational Alerts vs. Governance & Control Assurance Issues

i. Technical Operational Alerts (SOC / Sysadmin Focus)

These are real-time, actionable security events that require immediate investigation, containment, or triage by technical staff.

Examples from the lab:

- A rule trigger from auditd showing write access to sensitive files (-w /etc/passwd -p rwx -k passwd_changes).
- Multiple failed sudo password attempts for user icdfa within a short time window in auth.log.
- An unexpected change in auditd.service state (e.g., service stopped or disabled).

Target Audience: SOC Analysts, System Administrators, Incident Response Team.

Goal: Detect potential privilege escalation, unauthorized file manipulation, or brute-force attempts in real time.

ii. Governance & Control-Assurance Issues (GRC / Management Focus)

These represent systemic control failures, policy non-compliance, or baseline drift over time. They inform management whether controls remain effective and compliant with security frameworks (e.g., ISO 27001, CIS Benchmarks).

Examples from the lab:

- Control Effectiveness Failure: auditd being repeatedly stopped or missing required audit rules across endpoints, indicating a failure in Continuous Control Monitoring.
- Systemic Weakness: Lynis Hardening Index dropping below the organization's required compliance threshold (e.g., falling below 70/100 across a server fleet).
- Repeated Policy Violations: Persistent sudo authentication failures across multiple administrative accounts indicating an unaddressed training, policy, or identity management flaw.
- Configuration Drift: Disabling security monitoring components without an approved Change Management ticket.

Target Audience: Information Security Officer (CISO), Compliance Auditors, Risk Management Committee.

Goal: Validate regulatory compliance, track risk trends, allocate security budget, and update corporate policy.

Module 5 Applied Governance Task From Linux Evidence to Control Assurance

Control-Monitoring Table

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
Integrity Monitoring of Sensitive Account Files	Figure 3	IAM Team / System Administrator	Compliant	0 unauthorized writes to /etc/passwd or /etc/shadow	Mitigates local privilege escalation and creation of backdoor administrative accounts.	Restrict file write access to root and keep audit rules immutable	Re-query ausearch -k passwd_changes weekly.
Authentication & Access Rate Monitoring	Figure 5	SOC / Security Operations	Attention Required	>5 failed logins within a 10-minute window	Unblocked failure spikes indicate automated SSH brute-force password attacks.	Deploy Fail2ban to block bad IPs and enforce SSH key-based login.	Audit aureport -login and auth.log after 48 hours
Privileged Command Accountability	Figure 6	IT Audit / Security Assurance	Partially Compliant	100% of privileged commands tied to individual named accounts	Shared or unmonitored privileges obscure accountability during incident investigations	Restrict standard users in /etc/sudoers and enforce explicit session logging.	Review sudo usage logs monthly.
System Hardening & Configuration Baseline	Figure 11	Systems / Infrastructure Team	Needs Improvement	Minimum Lynis Hardening Index of 75/100	Weak kernel parameters and default configurations leave services vulnerable to exploit payloads.	Apply high-priority Lynis hardening suggestions (e.g., sysctl network parameters).	Run sudo lynis audit system post-remediation.
Audit Subsystem Uptime & Tamper Resistance	Figure 8	Systems Engineering	Active/Monitored	Auditd daemon uptime = 99.9%	Disabling or crashing auditd causes a total loss of system-level audit capability	Enable auditd service autostart and monitor service crashes in syslog.	Verify daemon status with auditctl -s bi-weekly

Governance Escalation Questions

1. Which finding is the highest priority and why?

Highest Priority: Unrestricted or recurring SSH authentication failures (auth.log) combined with sub-baseline system hardening.

Why: Remote authentication vector exposure represents an active, high-likelihood external threat that can lead to initial system access.

2. Who should own remediation for each major finding?

Authentication & Access Control Findings: Security Operations (SOC) & IAM Team.

OS Hardening & Patching Findings: Infrastructure & Systems Engineering.

Audit & Logging Coverage Findings: Security Assurance & Compliance.

3. What threshold or condition should trigger escalation to the CISO, Risk function, or management committee?

- Any unauthorized write event recorded against /etc/passwd or /etc/shadow.
- Disruption or termination of the auditd logging daemon lasting longer than 15 minutes.
- An overall Lynis Hardening Index score dropping below the mandatory risk threshold (<60/100).

4. What evidence would demonstrate successful remediation?

- Clean auth.log records showing blocked brute-force IPs after Fail2ban deployment.
- A updated Lynis audit report (lynis-report.dat) demonstrating an increased Hardening Index score and resolution of red-level warnings.

5. When and how should the control be retested?

Timeline: Within 14 business days following remediation implementation.

Method: Re-run automated Lynis scans (sudo lynis audit system), query auditd key logs (ausearch -k), and execute verification log searches via journalctl.